

1. (4 pts) How does changing the value of the load resistance of a photodetector affect the amplitude of the signal?

2. (4 pts) Sketch a circuit for a photoconductive mode Si PIN detector connected to the input of an oscilloscope. Also sketch a circuit for a Si PIN detector in a photovoltaic arrangement.

3. (3 pts) Assume that only junction capacitance limits the speed of a chosen PIN photodetector. If the junction resistance is 10^6 ohms and the load resistance is 50 ohms, what is the maximum junction capacitance that can be tolerated if you need to measure a laser pulse with a 10ns rise time?

4. (3 pts) How did the signal from the Si PIN detector compare to the signal from the Phototransistor. Draw the waveforms and explain. Which one is fastest?